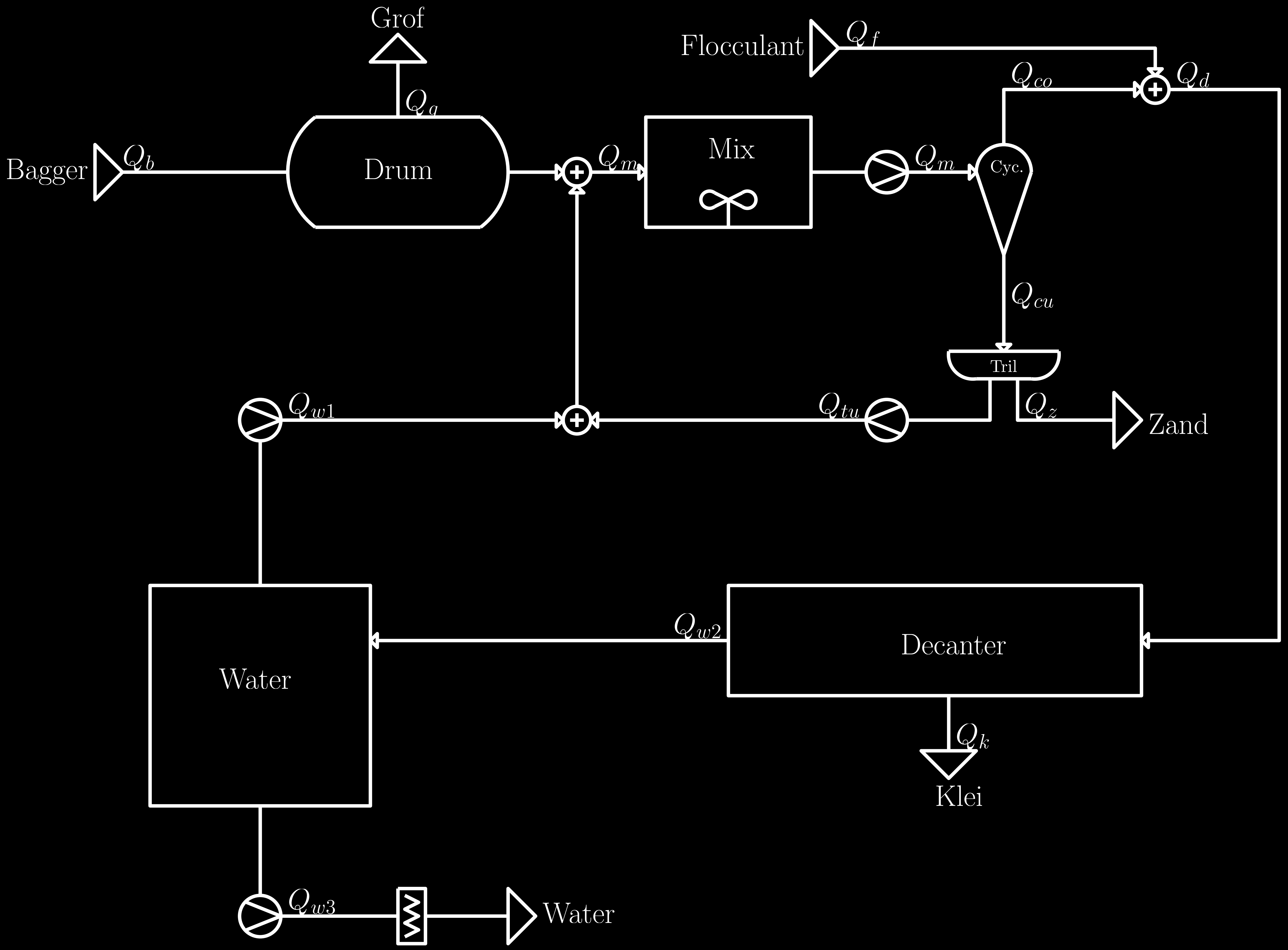
Advanced Cyclone Control

Waarom?

- Bagger is divers
  - Automatische adaptatie op baggersoort
- Betere scheiding
  - Onbemande proces optimalisatie

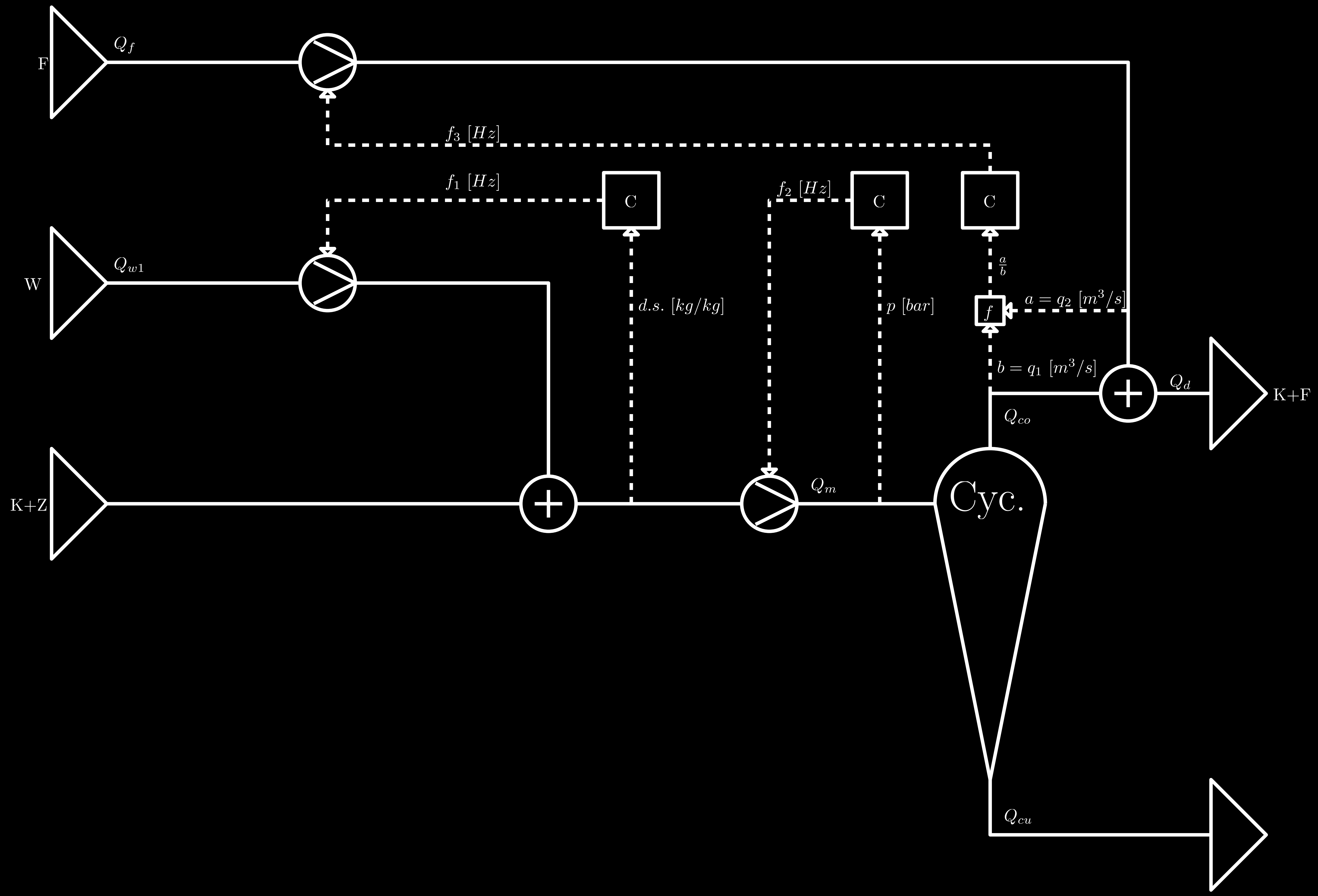
Decanter systeem



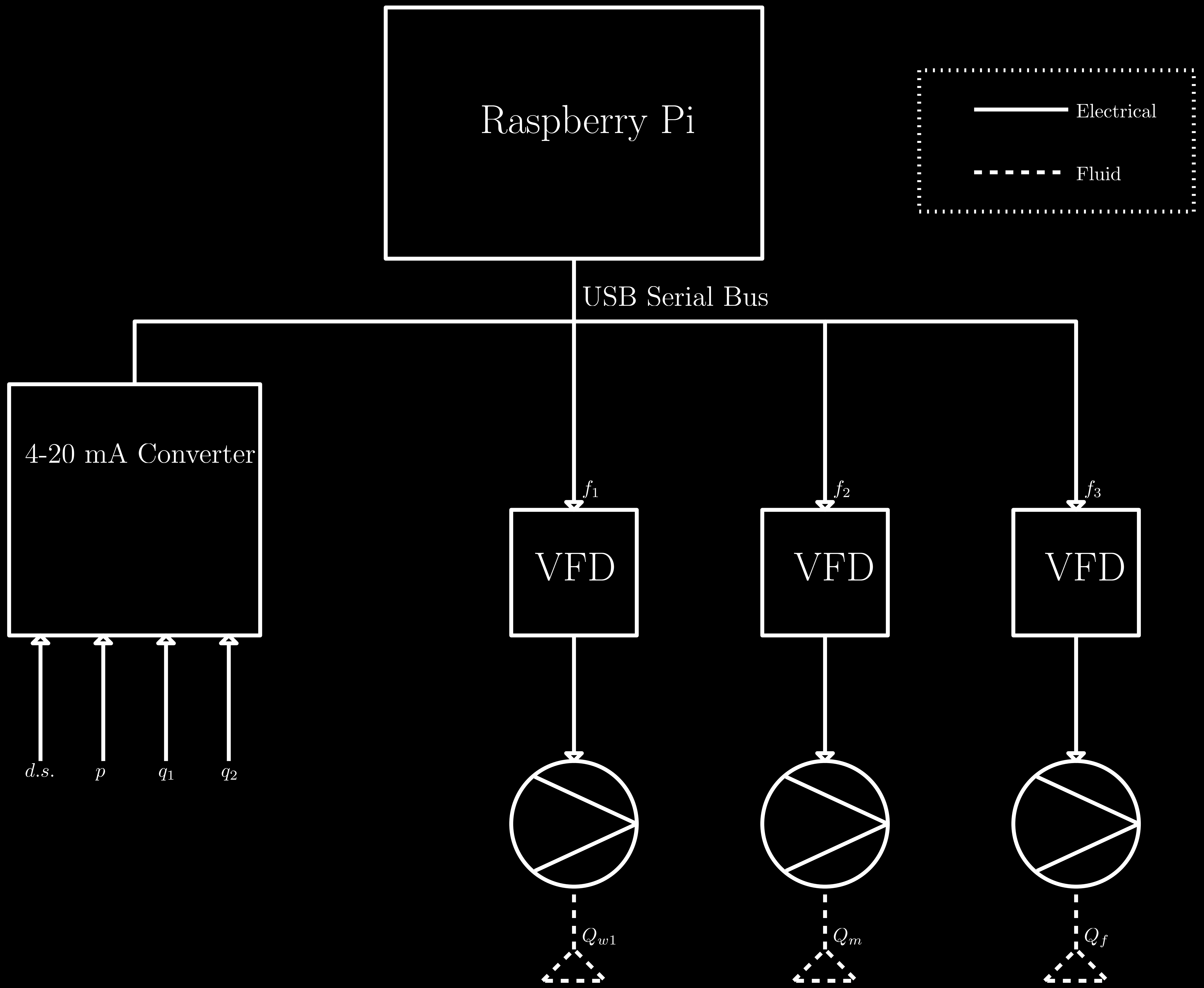
Hydrocycloon controle

- 3 feedback loops:
  - Ingangsdruk hydrocycloon [bar]
  - Droge stof gehalte ingang hydrocycloon [kg/kg]
  - Flocculant toevoeging uitgang hydrocycloon [L/L]

Control system



Implementatie

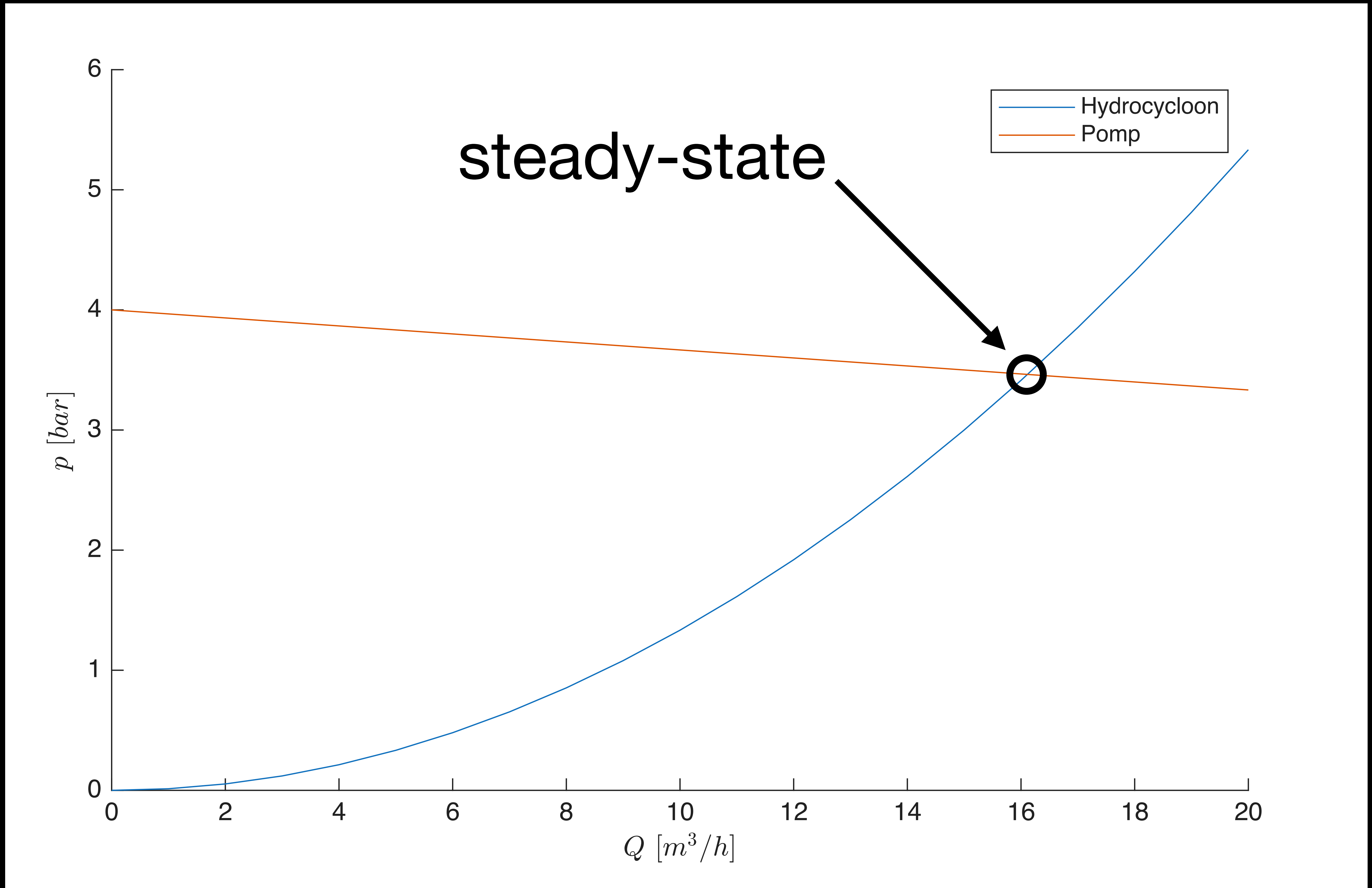


Plan

- Testopstelling iteratief
- 3 iteraties:
  - Druk
  - Drogestof
  - Flocculant

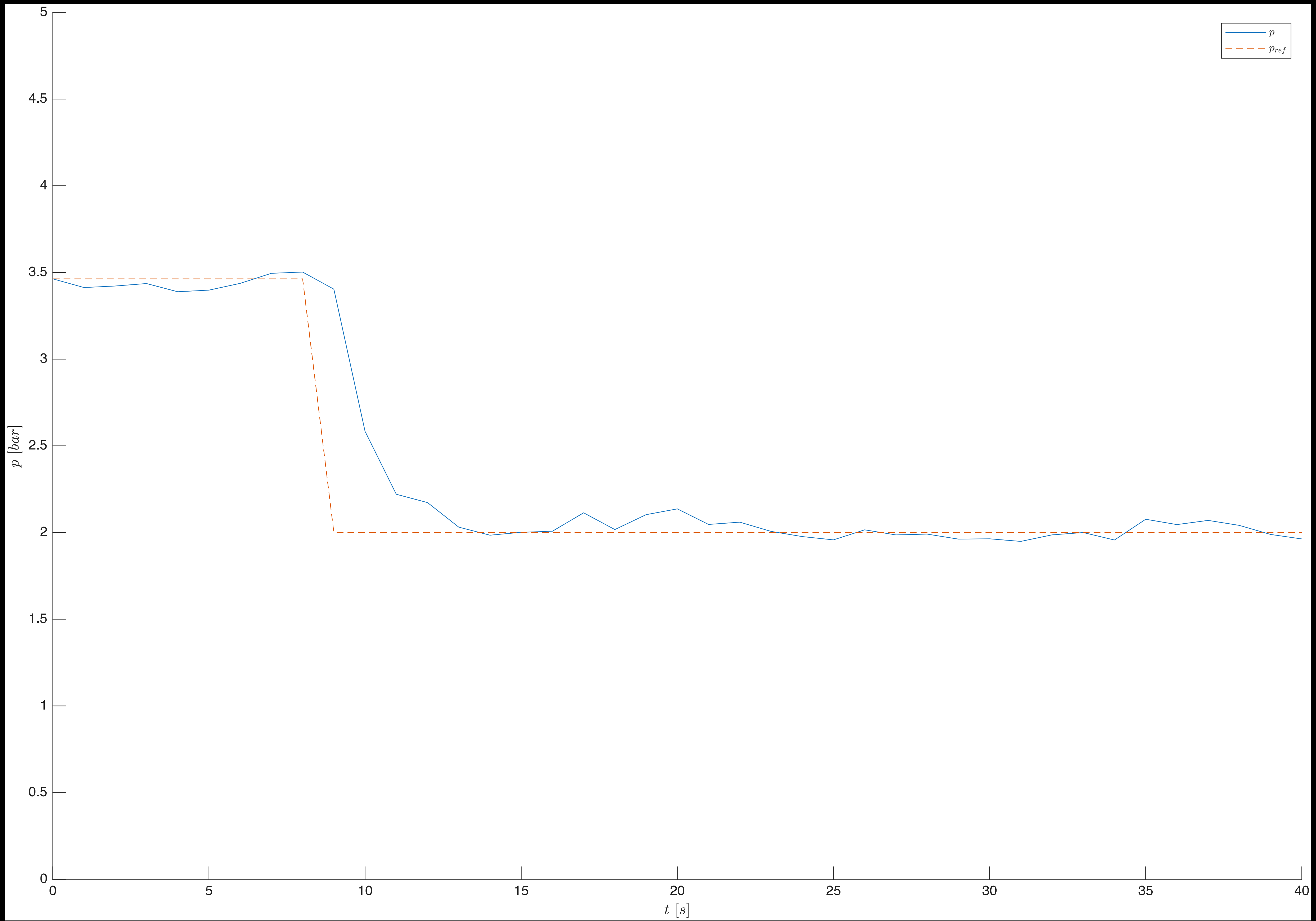
Druk

- Drukverlies hydrocycloon: $p_c = KQ^2$
- Drukopbouw pomp: $p_p \approx aQ + b$
- Steady-state: $p_c = p_p \quad \Rightarrow \quad Q_{ss}, p_{ss}$
- Pompfrequentie:
  - $\dfrac{Q_1}{Q_0} = \dfrac{f_1}{f_0}, \dfrac{p_1}{p_0} = \left(\dfrac{f_1}{f_0}\right)^2$
  - $p_p = aQ\dfrac{f_1}{f_0} + b\left(\dfrac{f_1}{f_0}\right)^2$

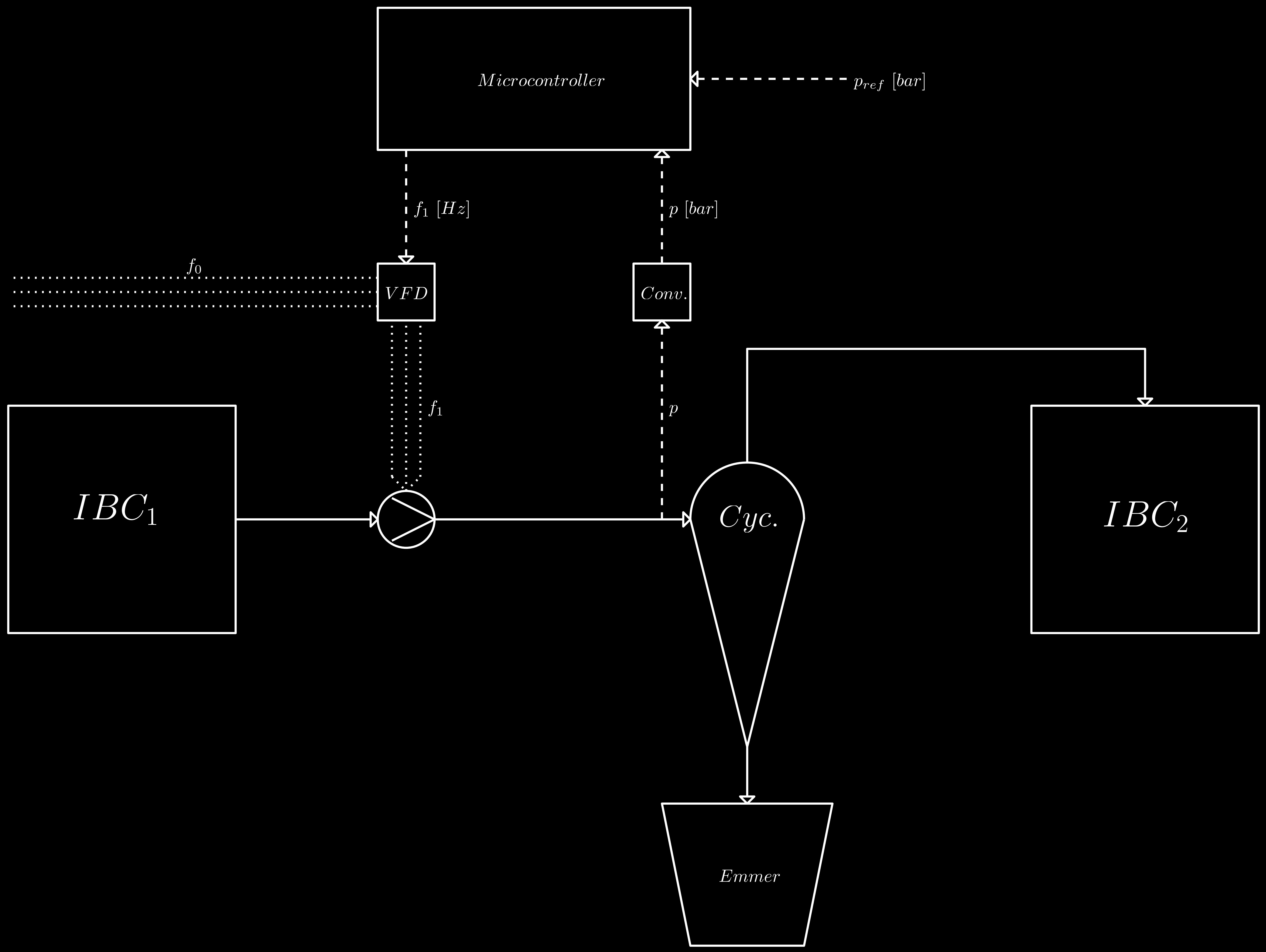


Simulatie

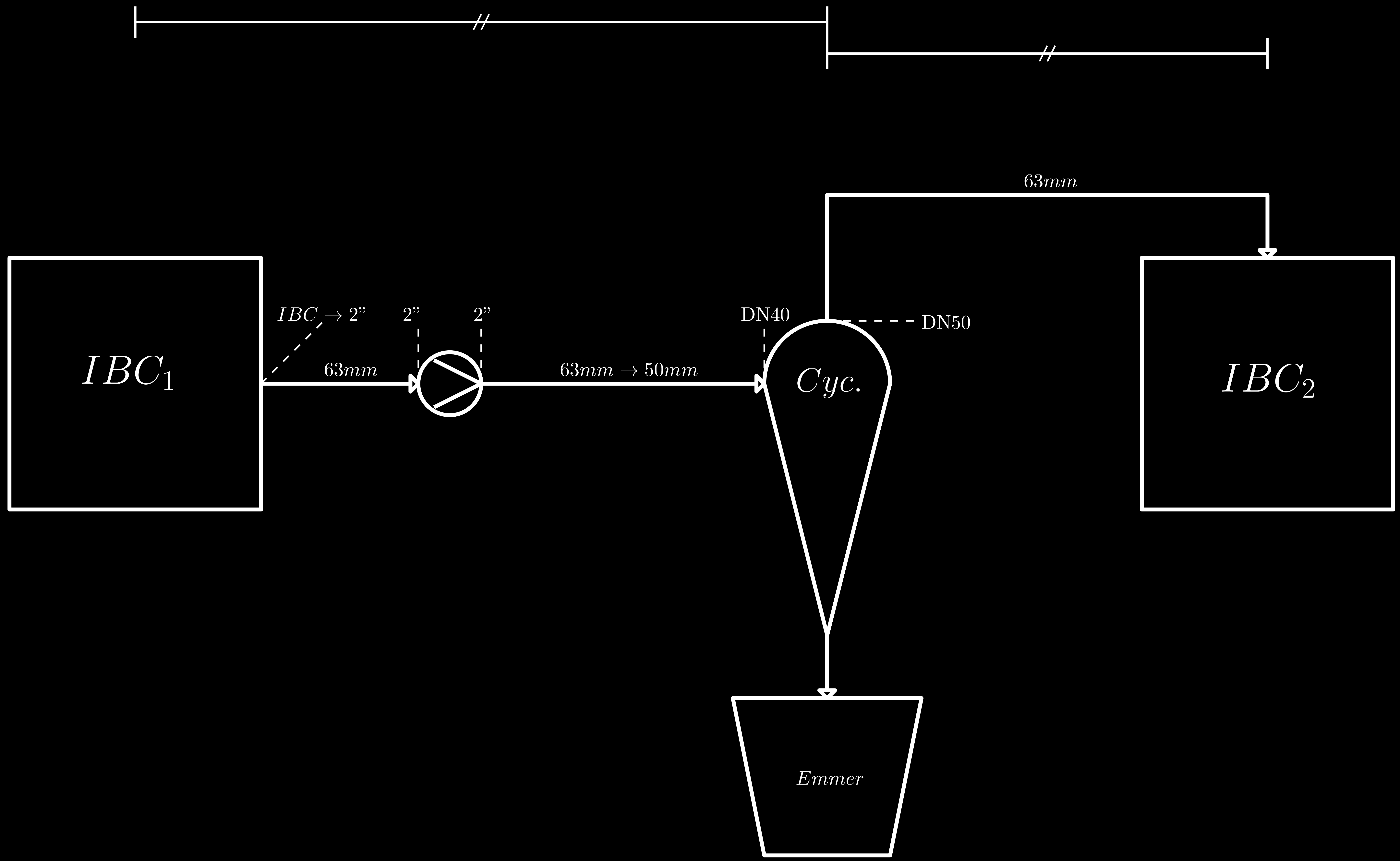
- MATLAB
- 2nd-order plant + ruis ($\sigma = 0.02$):
  - $\tau_p = 0.5\ [s], \tau_f = 1\ [s]$
  - $\Delta t = 1\ [s]$
- PD-controller:
  - $K_p = 8$
  - $K_d = 15$



Testopstelling



Buisvoering



Resultaat

- Controller parameter afstelling
- System response water
- System response bagger (< 2mm)